

Proof of the Four-Element Pairwise-Independent Correlation Gap

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Abstract

We prove, subject to an exact finite certificate described below, that for every nonnegative monotone submodular function on a four-element ground set and every real vector of singleton marginals, the unrestricted concave closure is at most $4/3$ times its pairwise-independent analogue. The constant is sharp. The finite premise is checked with exact rational arithmetic and replayed by a separate Node.js/**BigInt** implementation.

1 Proof of the four-element theorem

1.1 Statement and exact negation

Let $E = [4]$. For a probability law p on 2^E , write $p_S = \Pr_p(S)$. Given $x \in [0, 1]^4$, define

$$\mathcal{D}(x) = \left\{ p \in \mathbb{R}_{\geq 0}^{2^E} : \sum_{S \subseteq E} p_S = 1, \quad \sum_{S \ni i} p_S = x_i \quad (i \in E) \right\}.$$

Let $\mathcal{D}_{\text{PI}}(x)$ be the subset additionally satisfying

$$\sum_{S \supseteq \{i, j\}} p_S = x_i x_j \quad (i < j).$$

For a nonnegative monotone submodular set function $f : 2^E \rightarrow \mathbb{R}_{\geq 0}$, put

$$f^+(x) = \max_{p \in \mathcal{D}(x)} \mathbb{E}_p f, \quad f^{\text{PI}}(x) = \max_{q \in \mathcal{D}_{\text{PI}}(x)} \mathbb{E}_q f.$$

Theorem 1.1 (Four-element pairwise-independent correlation gap). *For every nonnegative monotone submodular $f : 2^{[4]} \rightarrow \mathbb{R}_{\geq 0}$ and every real $x \in [0, 1]^4$,*

$$3f^+(x) \leq 4f^{\text{PI}}(x). \tag{CG4}$$

The factor $4/3$ is best possible.

The exact negation of (CG4) is the existence of admissible (f, x) and $p \in \mathcal{D}(x)$ such that

$$3\mathbb{E}_p f > 4\mathbb{E}_q f \quad \text{for every } q \in \mathcal{D}_{\text{PI}}(x). \tag{NEG}$$

The universal quantifier over q is essential.

1.2 Normalization

Lemma 1.2 (Normalization). *It is enough, in both the proof and disproof directions, to consider functions satisfying $f(\emptyset) = 0$.*

Proof. Put $c = f(\emptyset) \geq 0$ and $g = f - c$. Monotonicity gives $g \geq 0$, and subtracting a constant preserves monotonicity and submodularity. Every expectation shifts by c , so

$$f^+ = c + g^+, \quad f^{\text{PI}} = c + g^{\text{PI}}. \quad (2.1)$$

If (CG4) holds for g , then

$$c + g^+ \leq c + \frac{4}{3}g^{\text{PI}} \leq \frac{4}{3}(c + g^{\text{PI}}).$$

Conversely, a strict violation by f gives

$$g^+ > \frac{4}{3}g^{\text{PI}} + \frac{c}{3} \geq \frac{4}{3}g^{\text{PI}}.$$

Thus normalization loses neither direction. $\square$

Let C_4 be the cone of normalized monotone submodular functions on E . If $g \in C_4$ and $g(E) = 0$, monotonicity forces $g = 0$. Every nonzero member can therefore be scaled uniquely into

$$\mathcal{F} = \{g \in C_4 : g(E) = 1\}.$$

This is a nonempty compact convex polytope, since $0 \leq g(S) \leq 1$ for every $S \subseteq E$.

1.3 The simultaneous-dominance reduction

Lemma 1.3 (Simultaneous dominance). *For fixed x , (CG4) for all normalized functions is equivalent to the following statement: for every $p \in \mathcal{D}(x)$, there is one $q \in \mathcal{D}_{\text{PI}}(x)$ such that*

$$4\mathbb{E}_q g \geq 3\mathbb{E}_p g \quad (g \in C_4). \quad (3.1)$$

Proof. Fix p , let $Q = \mathcal{D}_{\text{PI}}(x)$, and define

$$L(g, q) = 3\mathbb{E}_p g - 4\mathbb{E}_q g \quad (g \in \mathcal{F}, q \in Q).$$

The product law lies in Q , so Q is nonempty; it is also compact and convex. Finite-dimensional bilinear minimax gives

$$\max_{g \in \mathcal{F}} \min_{q \in Q} L(g, q) = \min_{q \in Q} \max_{g \in \mathcal{F}} L(g, q). \quad (3.2)$$

For fixed g , the inner minimum on the left is

$$3\mathbb{E}_p g - 4g^{\text{PI}}(x).$$

If (CG4) holds, this is nonpositive for every p and g , because $\mathbb{E}_p g \leq g^+(x)$. Equation (3.2) then supplies a single q for which $L(g, q) \leq 0$ for every $g \in \mathcal{F}$. Homogeneity extends the result from $\mathcal{F}$ to C_4 , including the zero function.

Conversely, apply (3.1) to a maximizer p of $g^+(x)$. Then

$$3g^+(x) \leq 4\mathbb{E}_q g \leq 4g^{\text{PI}}(x).$$

$\square$

This is the reason that one may use a common finite ray test. Choosing a different q for each ray would not prove (3.1).

1.4 Extreme unrestricted laws and sorting

Let A be the 5×16 first-moment matrix with columns

$$a(S) = (1, \mathbf{1}_{1 \in S}, \dots, \mathbf{1}_{4 \in S})^T.$$

It has rank five. Indeed, the columns indexed by $\emptyset, \{1\}, \dots, \{4\}$ are linearly independent.

Lemma 1.4 (Extreme-law reduction). *It is enough to prove (3.1) for extreme points of $\mathcal{D}(x)$. Every such point has an affinely independent positive support of size at most five, and every smaller support extends to a five-column basis of A .*

Proof. If $p = \sum_t \alpha_t p^{(t)}$ is a convex combination of extreme laws and $q^{(t)}$ simultaneously dominates $p^{(t)}$, then $q = \sum_t \alpha_t q^{(t)}$ has the same mass, first moments, and pair moments as every $q^{(t)}$. Hence $q \in \mathcal{D}_{\text{PI}}(x)$, and linearity gives (3.1).

If the columns $a(S)$ in the positive support of an extreme p were linearly dependent, a nonzero supported vector h would satisfy $Ah = 0$. For sufficiently small $\varepsilon > 0$, both $p + \varepsilon h$ and $p - \varepsilon h$ would be distinct feasible laws, contradicting extremality. Thus the support columns are independent and there are at most five. The basis-extension theorem extends any independent subset of columns to a five-column basis. Assigning zero weights to the added masks realizes the original law on a face of that basis simplex. $\square$

There are exactly 3,008 such five-column bases; this is independently recomputed by exact rank enumeration. All objects are equivariant under a simultaneous permutation of the four ground elements. We may therefore relabel x, p , apply the construction, and pull the resulting q back. Hence it is enough to consider

$$0 \leq x_1 \leq x_2 \leq x_3 \leq x_4 \leq 1. \quad (4.1)$$

Ties cause no difficulty because every region below is closed.

1.5 Two analytic regions

1.5.1 Low total marginal

Lemma 1.5 (Low-total region). *If $s = \sum_i x_i \leq 1$, then (CG4) holds.*

Proof. Assume (4.1). Prescribe candidate intersection moments by $m_i = x_i$, $m_{ij} = x_i x_j$, and, for $|T| \geq 3$, let m_T be the product of the two smallest marginals indexed by T . Möbius inversion gives the following potentially nonzero atom masses; all unlisted masses are zero:

T	q_T	
$\emptyset$	$1 - s + x_4(s - x_4)$	
$\{i\}, i = 1, 2, 3$	$x_i(1 - x_4)$	
$\{4\}$	$x_4(1 - s + x_4) + x_3(x_1 + x_2)$	(5.1)
$\{1, 4\}, \{2, 4\}$	$x_1(x_4 - x_3), \quad x_2(x_4 - x_3)$	
$\{3, 4\}$	$x_3(x_4 - x_1 - x_2) + x_1 x_2$	
$\{1, 3, 4\}, \{2, 3, 4\}$	$x_1(x_3 - x_2), \quad x_2(x_3 - x_1)$	
E	$x_1 x_2$	

The empty mass is nonnegative because $s \leq 1$. All other masses are manifestly nonnegative except possibly $q_{\{3,4\}}$, and

$$q_{\{3,4\}} \geq x_3(x_3 - x_1 - x_2) + x_1 x_2 = (x_3 - x_1)(x_3 - x_2) \geq 0.$$

The displayed nonnegative masses define a law Q ; equivalently, their Möbius transform is the prescribed moment vector and in particular has total mass one. The defining intersection moments show exactly that Q has marginals x_i and pair intersections $x_i x_j$, so it is pairwise independent.

For every $A \subseteq E$,

$$\Pr(Q \cap A \neq \emptyset) \geq \frac{3}{4} \sum_{i \in A} x_i. \quad (5.2)$$

For two coordinates with sum $r \leq 1$, this follows from $4uv \leq (u+v)^2 \leq u+v=r$. For three coordinates with sorted marginals $a \leq b \leq c$, inclusion–exclusion and the chosen triple moment give

$$\Pr(Q \cap A \neq \emptyset) = c + (a+b) - c(a+b),$$

and the same inequality applies because $c + a + b \leq 1$. For $A = E$, the union probability is $s - x_4(s - x_4)$, and the identical argument uses $c = x_4$, $a + b = s - x_4$. Singletons are immediate.

Normalize $f(\emptyset) = 0$, put $a_i = f(\{i\})$, and define $h(\emptyset) = 0$ and $h(T) = \max_{i \in T} a_i$ for nonempty T . Monotonicity gives $f \geq h$. Layer cake and (5.2) give

$$\begin{aligned} \mathbb{E}h(Q) &= \int_0^\infty \Pr(Q \cap \{i : a_i \geq u\} \neq \emptyset) du \\ &\geq \frac{3}{4} \sum_i x_i a_i. \end{aligned} \quad (5.3)$$

Normalized submodularity implies subadditivity and hence $f(T) \leq \sum_{i \in T} a_i$. Thus every law with marginals x has expected value at most $\sum_i x_i a_i$. The law with mass x_i on $\{i\}$ and mass $1-s$ on $\emptyset$ attains the bound, so

$$f^+(x) = \sum_i x_i a_i.$$

Together with (5.3), this proves $f^{\text{PI}}(x) \geq 3f^+(x)/4$. The argument uses no division and includes all zero and deterministic boundary cases. $\square$

1.5.2 A large marginal

We use the exact three-element theorem

$$4u^{\text{PI}}(y) \geq 3u^+(y) \quad (\text{N3})$$

for every normalized nonnegative monotone submodular u and every real $y \in [0, 1]^3$; see Ramachandra and Natarajan [1] and the exact campaign reconstruction.

Lemma 1.6 (Large-coordinate region). *If $x_i = t \geq 2/3$ for some coordinate, then (CG4) holds.*

Proof. Normalize $f(\emptyset) = 0$, put $a = f(\{i\})$, and for $T \subseteq E \setminus \{i\}$ define

$$G(T) = (1-t)f(T) + tf(T \cup \{i\}). \quad (5.4)$$

Here $T \mapsto f(T)$ is deletion of coordinate i , while $T \mapsto f(T \cup \{i\})$ is contraction. Deletion and contraction preserve nonnegativity, monotonicity, and submodularity; so does their convex combination. Also $G(\emptyset) = ta$, and monotonicity gives $G - G(\emptyset) \geq 0$. Applying (N3) to $G - G(\emptyset)$ yields the strengthened nonnormalized form

$$4G^{\text{PI}}(x_{-i}) \geq 3G^+(x_{-i}) + ta. \quad (5.5)$$

Adjoin to any three-coordinate PI law an independent Bernoulli- t bit at coordinate i . This is a four-coordinate PI law and its expected f -value is the expected G -value. Hence

$$f^{\text{PI}}(x) \geq G^{\text{PI}}(x_{-i}). \quad (5.6)$$

Let P attain $f^+(x)$, write X_i for its i -th bit and $Y = S \setminus \{i\}$, and set

$$d(Y) = f(Y \cup \{i\}) - f(Y).$$

Diminishing returns and monotonicity give $0 \leq d(Y) \leq a$. If only X_i is independently resampled with mean t , its expected value is

$$R = \mathbb{E}[(1-t)f(Y) + tf(Y \cup \{i\})] = \mathbb{E}G(Y) \leq G^+(x_{-i}). \quad (5.7)$$

Moreover,

$$\begin{aligned} f^+(x) - R &= \mathbb{E}[(X_i - t)d(Y)] \\ &\leq (1-t)\mathbb{E}[X_i d(Y)] \\ &\leq t(1-t)a. \end{aligned} \quad (5.8)$$

Combining (5.5)–(5.8),

$$4f^{\text{PI}}(x) \geq 3f^+(x) + ta(3t-2) \geq 3f^+(x).$$

This includes $t = 2/3$, $t = 1$, and $a = 0$. □

By Lemmas 1.5 and 1.6, after sorting it remains only to cover the closed central region

$$0 \leq x_1 \leq x_2 \leq x_3 \leq x_4 \leq \frac{2}{3}, \quad \sum_i x_i \geq 1. \quad (\text{C})$$

Using a closed superset is deliberate: the two analytic regions already cover the equality walls, and overlap creates no gap.

1.6 The finite normalized-function test

For $i \notin S$, write $S+i = S \cup \{i\}$; for distinct $i, j \notin S$, write $S+i+j = S \cup \{i, j\}$. Also write $\Delta_i g(S) = g(S \cup \{i\}) - g(S)$. The normalized cone C_4 is defined exactly by the four terminal marginal inequalities

$$g(E) - g(E \setminus \{i\}) \geq 0 \quad (\text{M})$$

and the 24 elementary diminishing-returns inequalities

$$g(S+i) + g(S+j) - g(S) - g(S+i+j) \geq 0 \quad (\text{DR})$$

for $i < j$ and $S \subseteq E \setminus \{i, j\}$. Indeed, (DR) telescopes to all diminishing-return inequalities and hence all submodularity inequalities; together with terminal monotonicity it also telescopes to every nonnegative marginal.

Proposition 1.7 (Finite cone certificate). *The pointed 15-dimensional cone cut out by (M) and (DR) has exactly 41 primitive extreme rays in 11 S_4 -orbits.*

Exact finite premise. The standard-library double-description transcript reconstructs the 28 rows, starts from a certified simplicial cone, inserts the remaining 13 halfspaces, and ends with 41 rays. Rank-14 integer minors certify every final ray and every facet. The cone-certificate SHA-256 is

4d99cba4266f6a9afbbae04a9fff7a411cf450dfd78fd261d222a6d3a9d5ba13d,

and the primitive-ray SHA-256 is

9f0f3d8e71de639468ea5915a4adfdff4056321040c7221def886e201624eff1.

The verifier reconstructs the defining rows and replays each double-description transition; the certificate is therefore a completeness certificate, not merely a supplied feasible ray list.

For fixed p, q , conic generation makes (3.1) equivalent to its 41 ray inequalities, with the same q in every inequality. Ten rays are the coverage functions

$$h_A(S) = \mathbf{1}[S \cap A \neq \emptyset], \quad 1 \leq |A| \leq 2. \quad (6.1)$$

Their inequalities are automatic for every $p \in \mathcal{D}(x)$ and every $q \in \mathcal{D}_{\text{PI}}(x)$. For $A = \{i\}$, the slack is x_i . For $A = \{i, j\}$, write $u = x_i$, $v = x_j$, and $s = u + v$. Then

$$\mathbb{E}_q h_A = s - uv, \quad \mathbb{E}_p h_A \leq \min(1, s).$$

If $s \leq 1$, then $uv \leq s^2/4 \leq s/4$. If $s \geq 1$, put $a = 1 - u$, $b = 1 - v$; then $a + b \leq 1$, $ab \leq 1/4$, and $s - uv = 1 - ab \geq 3/4$. Thus only 31 ray inequalities remain for the central certificate.

1.7 The exact central certificate

Proposition 1.8 (Central certificate; computed exactly). *For every five-column basis B , every λ in its closed standard simplex whose induced marginal vector lies in (C), and the corresponding unrestricted law p , at least one of the eight recipes below is a law $q \in \mathcal{D}_{\text{PI}}(x)$ whose eleven atom masses and all 31 nonautomatic simultaneous ray slacks are nonnegative.*

The remainder of this subsection specifies the finite object and the exact checks establishing Proposition 1.8.

Encode subsets by masks $0, \dots, 15$, with bit $i - 1$ representing element i . Fix a five-mask first-moment basis $B = (B_0, \dots, B_4)$ and barycentric weights

$$\lambda_k \geq 0, \quad \sum_{k=0}^4 \lambda_k = 1, \quad p_{B_k} = \lambda_k. \quad (7.1)$$

Then

$$x_i = \sum_k \lambda_k \mathbf{1}_{i \in B_k}. \quad (7.2)$$

1.7.1 Eight explicit PI recipes

Let

$$m(x) = (1, x_1, x_2, x_3, x_4, x_1x_2, x_1x_3, x_1x_4, x_2x_3, x_2x_4, x_3x_4)^T. \quad (7.3)$$

For an 11-mask set R , let M_R be the 11×11 matrix whose mask column records mass, four singleton indicators, and six pair indicators. The eight recipe supports are

$R_0 = (0, 3, 4, 6, 8, 9, 10, 12, 13, 14, 15)$
 $R_1 = (1, 4, 6, 7, 8, 10, 11, 12, 13, 14, 15)$
 $R_2 = (1, 2, 4, 7, 8, 9, 10, 12, 13, 14, 15)$
 $R_3 = (0, 1, 2, 4, 8, 9, 10, 12, 13, 14, 15)$
 $R_4 = (0, 1, 2, 3, 4, 5, 6, 9, 10, 12, 15)$
 $R_5 = (1, 2, 3, 4, 5, 8, 9, 10, 12, 14, 15)$
 $R_6 = (1, 3, 5, 6, 7, 8, 10, 11, 12, 13, 15)$
 $R_7 = (0, 1, 2, 4, 5, 8, 9, 10, 12, 14, 15)$

Every M_R is exactly invertible. Recipe R is the law supported on R with atom vector

$$q_R(x) = M_R^{-1}m(x). \quad (7.4)$$

If its eleven entries are nonnegative, equations (7.3)–(7.4) make it a probability law with the required first and pair moments. For each of the 31 nonautomatic cone rays r , impose

$$4 \sum_{S \in R} q_{R,S}(x)r(S) - 3 \sum_{k=0}^4 \lambda_k r(B_k) \geq 0. \quad (7.5)$$

Thus a recipe has 42 constraints: eleven atom inequalities and 31 dominance inequalities. Since x is affine in λ , every constraint is a rational polynomial of degree at most two in λ .

1.7.2 Closed chamber cover

Inside the standard λ -simplex impose the homogeneous linear forms equivalent to (C). Split first on

$$T = x_1 + x_2 + x_3 - 1.$$

The cell $T \leq 0$ is one chamber. On $T \geq 0$, split independently on

$$P = x_1 + x_2 - 1, \quad H = 1 + x_1 - x_2 - x_3.$$

This gives four more chambers. Because both signs are non-strict, these five closed chambers cover the whole central cell, including all walls.

Two basis-specific refinements are used, always with both closed signs:

$$B=(1,6,8,10,12): \quad 2 \lambda_0 - \lambda_1 - \lambda_3 = 0;$$

$B=(2,5,8,9,12)$, in the $T \geq 0$, $P \leq 0$, $H \geq 0$ chamber:

$$\begin{aligned} \lambda_0 + \lambda_1 - \lambda_2 + \lambda_3 - \lambda_4 &= 0, \\ -\lambda_0 + \lambda_1 - \lambda_2 - \lambda_3 + \lambda_4 &= 0. \end{aligned}$$

The latter two forms equal, respectively, $2(x_1 + x_2 - 1/2)$ and $2(x_3 - 1/2)$. Both-sign splitting preserves coverage even on their common zero sheets.

For each basis and sign chamber, exact active-set enumeration lists every vertex of the bounded rational polytope. A point is a vertex exactly when four active inequality normals, together with $\sum \lambda = 1$, have full rank; hence enumerating all four-row choices is complete. Every nonempty bounded cell has a vertex. Exact pulling triangulation then covers the cell: recursively choose the least vertex and cone it over pulling triangulations of all facets not containing it. Facets are recovered from the original inequalities by exact affine-rank tests.

1.7.3 Exact quadratic checks on a simplex

Let $v_0, \dots, v_d$ be a simplex and write $\lambda = \sum_i y_i v_i$, where $y_i \geq 0$ and $\sum_i y_i = 1$. A quadratic constraint h has the exact representation

$$h(\lambda) = y^T H y. \quad (7.6)$$

The diagonal controls are $h(v_i)$, and the off-diagonal controls are

$$H_{ij} = 2h\left(\frac{v_i + v_j}{2}\right) - \frac{h(v_i) + h(v_j)}{2}. \quad (7.7)$$

These are the degree-two Bernstein controls; (7.7) is an identity, not a positivity heuristic.

Lemma 1.9 (Quadratic simplex lemma). *Exact enumeration of every face-stationary affine set decides whether (7.6) is nonnegative on the whole simplex.*

Proof. If h is negative, its compact-simplex minimum is negative. On the relative interior of its support face I , Lagrange stationarity gives

$$(Hy)_i = (Hy)_j \quad (i, j \in I), \quad \sum_{i \in I} y_i = 1. \quad (7.8)$$

The exact checker intersects (7.8) with $y \geq 0$ by enumerating basic feasible solutions. If y, z satisfy the same stationary system, then $Hy = \alpha \mathbf{1}$ and $H z = \beta \mathbf{1}$ on I , so symmetry gives

$$\alpha = z^T H y = y^T H z = \beta.$$

Consequently $y^T H y = \alpha = \beta = z^T H z$: the quadratic value is constant on the stationary affine set. Testing one exact feasible basic point per nonempty stationary set therefore detects a negative minimum if and only if one exists. This also handles singular stationary systems and boundary minima. $\square$

1.7.4 Accepted simplex labels

For every simplex the certificate records one of the following exact labels.

1. **Ordinary recipe.** All 42 constraints of one recipe are nonnegative on the simplex by Lemma 1.9.
2. **Complementary switch.** Recipe R has at most one uncertified constraint u , recipe R' has at most one uncertified constraint v , and $v = -cu$ identically for an exact rational $c > 0$. If $u \geq 0$, the first recipe works; if $u \leq 0$, the second works. At $u = 0$, both blockers vanish. The recurring global identity is

$$q_{R_7, \emptyset} = -2q_{R_5, \{1, 2\}},$$

where mask 3 represents $\{1, 2\}$.

3. **Farkas/disjunctive label.** For each selected recipe, all constraints except a listed blocker set are nonnegative. For every selection of one blocker h_j from each selected recipe, the certificate gives exact weights $c_j \geq 0$, not all zero, such that

$$\sum_j c_j h_j \geq 0 \quad (7.9)$$

throughout the simplex, again by Lemma 1.9. If all selected recipes failed at one point, choose a strictly negative blocker from each. The left side of (7.9) would be strictly negative, a contradiction. The one-blocker-per-recipe special case is called a Farkas label.

The replay verifier recomputes all non-blocker constraints; it does not trust exploratory safety flags.

1.7.5 Retained exact certificate

The retained certificate covers all 3,008 bases and has the following exact totals:

closed chamber/sign records	15,048
nonempty records	9,288
empty records	5,760
initial simplices/leaves	17,933
ordinary-recipe leaves	17,906
switch leaves	21
Farkas/disjunctive leaves	6
maximum subdivision depth	0

The canonical payload is `verification/central/central-cell-cover-v1.json.gz`. Its sizes and hashes are

raw JSON bytes	1,352,318
gzip bytes	115,720
raw JSON SHA-256	c18f2096019a80ec495209863bdd8e46faa8e9e812c11e014cfc334b2a02f7ec
gzip SHA-256	dbad1fcaee0e0285b448e1179c73ea747b5225c52dbd60da70064a1cc6727958

The exporting scan used exact `Fraction` arithmetic. A standalone standard-library Python bundle reconstructed and replayed all four ranges, including 286,253 label-local exact quadratic checks. A separate Node.js implementation using only built-ins and custom `BigInt` rationals independently reconstructed the 3,008 bases, every chamber H-polytope, all vertices and pulling simplices, the eight recipe inverses and 42 constraints, and all label semantics. It performed 754,395 exact face-stationary quadratic tests and found no discrepancy. Thus the simplex argument covers all real barycentric points, not merely rational or sampled points.

1.8 Assembly of the proof

Proof of Theorem 1.1. First normalize f using Lemma 1.2. Fix a real $x \in [0, 1]^4$. If $\sum_i x_i \leq 1$, Lemma 1.5 proves (CG4). If $\max_i x_i \geq 2/3$, Lemma 1.6 proves it.

Otherwise fix an extreme $p \in \mathcal{D}(x)$, sort the coordinates, and relabel p accordingly. Then x belongs to the closed central region (C). By Lemma 1.4, the support of p extends with zero weights to one of the 3,008 five-column bases. Its barycentric vector lies in at least one of the closed chamber/sign cells and then in at least one simplex of that cell’s pulling triangulation. The simplex label supplies at least one recipe whose eleven atom probabilities and 31 nonautomatic dominance slacks are all nonnegative. The ten remaining ray slacks are automatic. Thus one $q \in \mathcal{D}_{\text{PI}}(x)$ satisfies all 41 ray inequalities and hence simultaneously dominates p on all of C_4 .

Convexly combine these q ’s to handle an arbitrary $p \in \mathcal{D}(x)$. Lemma 1.3 then gives (CG4) for every normalized function. Lemma 1.2 restores arbitrary $f(\emptyset) \geq 0$, and undoing the coordinate permutation restores the original labeling. This covers all real marginals, zero barycentric weights, chamber walls, the equality cases $\sum x_i = 1$ and $\max x_i = 2/3$, and all ties among sorted coordinates.

For sharpness, take the two-element OR function embedded on four elements and

$$x = (1/2, 1/2, 0, 0).$$

An unrestricted law may choose exactly one of the first two elements, so $f^+(x) = 1$. Pairwise independence fixes their intersection at $1/4$, so their union has probability $3/4$ and $f^{\text{PI}}(x) = 3/4$. Hence the ratio is $4/3$. $\square$

1.9 Scope and residual trust

The argument is computer-assisted. Its finite premises use exact rational arithmetic rather than floating-point evidence, and the decisive central certificate has a second-language replay. The residual trust base consists of the elementary polyhedral and quadratic lemmas stated above, the retained certificate bytes, JSON/gzip parsing, and the correctness of the standard-language run-times and exact-arithmetic implementations. This is not a Lean, Coq, or Isabelle formalization. The appropriate status is *candidate resolution pending expert review*.

References

- [1] A. Ramachandra and K. Natarajan, *Pairwise independent correlation gap*, Operations Research Letters **60** (2025), Article 107255. doi:10.1016/j.orl.2025.107255.